TEACHER GUIDE

High school business

Profit margins



A review lesson. Students run a car dealership in firms of four and review profit margins on their own deals. Seat each firm together. Each firm answers the topic tasks together.

The lesson

45-minute setting

Setup	Round 1	Round 2	Round 3	Final questions

EVERY ROUND

Firm decisions
3 min

→

Deals
4½–5 min

→

Topic tasks
4 min

→

B2B deal
2½ min
from Round 2

LESSON SETTINGS

30 min | **45 min** | 60 min | 75 min

What students do

Every student buys and sells cars. In Firm decisions, each role has its own job.



CEO

Staff · Sabotage from Round 2



Finance

Capital & Loans · Investments



Sales & Marketing

Marketing



Operations

Seasonal inventory

Topic tasks

Each firm gets its own figures. The method is the same. Your Live screen shows each task and every firm's answer.

1 Gross profit margin

Round 1

Last year's books.

Gross profit margin, as a %?

Cars sold	**12 cars**
Price per car	**$16,500**
Gross profit	**$45,540**

ANSWER

23%

12 × $16,500 = $198,000; $45,540 ÷ $198,000 × 100 = 23%

Why: It keeps 23 cents of every $1 of sales before overhead.

2 Why net margin fell

Round 1

Gross profit margin stayed the same, but net profit margin fell.

Which two could explain this?

- **A** Cars cost more to buy
- **B** Prices were cut
- **C** Staff wages rose
- **D** More cars were sold
- **E** Rent rose

ANSWER

✓ Staff wages rose
✓ Rent rose

Why: Only overhead comes between gross and net profit.

3 Rank gross margin

Round 2

Three dealers: sales and gross profit for the year.

Order by gross margin, highest first?

- **A** Sales $200,000, gross profit $50,000
- **B** Sales $80,000, gross profit $24,000
- **C** Sales $500,000, gross profit $100,000

ANSWER

1. **Sales $80,000, gross profit $24,000**
2. **Sales $200,000, gross profit $50,000**
3. **Sales $500,000, gross profit $100,000**

Why: Margins 30%, 25%, 20%: the biggest profit has the lowest margin.

Topic tasks

4 Gross margin change

Round 2

Two years compared.

Gross margin change, in % points?

Sales, last year	**$200,000**
Gross profit, last year	**$32,000**
Sales, this year	**$190,000**
Gross profit, this year	**$47,500**

ANSWER

9%

$47,500 ÷ $190,000 = 25%;
$32,000 ÷ $200,000 = 16%;
25 − 16 = 9 points

Why: The margin went from 16% to 25%.

5 Raise gross margin

Round 3

You want a higher gross profit margin.

Which change works, and why?

A Buy cars more cheaply

B Cut the showroom rent

ANSWER

✔ Buy cars more cheaply

Why: Rent is an overhead; gross margin is price less the cost of the cars.

6 New net margin

Round 3

Overhead is going up.

New net profit margin, as a %?

Sales	**$200,000**
Gross profit	**$19,900**
Overhead	**$9,000**
Overhead rises by	**10%**

ANSWER

5%

$9,000 + 10% = $9,900;
$19,900 − $9,900 = $10,000;
÷ $200,000 = 5%

Why: Work out the new overhead first, then the margin.

Final questions

Each student answers 6 final questions. They come from these 10 questions on cash, profit and sales tax.

1. You sell an EV city car to a customer for $11,000. In Indiana the customer pays 7% sales tax on top of the price. How much cash comes into your firm from this sale?
   ✔ **$11,770**

2. This round your firm sold cars to customers for $10,000, $17,000 and $21,000, and sold a Forklift to another dealer for $14,000. How much sales tax does the round's return ask for?
   ✔ **$3,360**

3. Your cash is $100,000 and your firm owes $2,100 of sales tax it has not filed yet. How does this affect the firm's net position?
   ✔ **It is $2,100 lower: unpaid sales tax is a debt and is subtracted**

4. Who pays the 7% sales tax on a car your firm sells to a customer?
   ✔ **The customer pays it on top of the price; your firm collects it and pays it to the Indiana Department of Revenue**

5. You bought a Panel van for $15,000 and sell it to a customer for $18,000. What is your gross profit on the van?
   ✔ **$3,000**

6. Your firm misses the round's sales tax return with $1,800 due. The late charge is 10% of the tax due, at least $5. What happens?
   ✔ **A $180 charge, and the $1,800 is still owed**

7. Cash $40,000, inventory at cost $30,000, investments $10,000, loans $25,000 and unpaid sales tax $1,400. What is the net position?
   ✔ **$53,600**

8. Why can a firm's cash look higher than the money it can really keep?
   ✔ **It holds sales tax collected from customers that belongs to the state until the return is filed**

9. A customer buys two Scooters 125cc at $3,900 each. How much sales tax does the customer pay?
   ✔ **$546**

10. Your firm collected $2,450 of sales tax this round and files the return on time. Cash before filing is $52,000. What is the cash after filing?
    ✔ **$49,550**